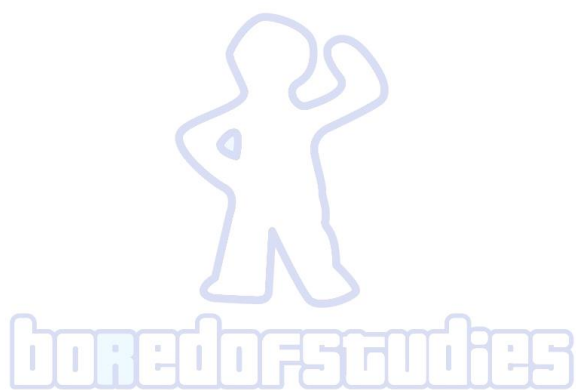




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2020

BORED OF STUDIES TRIAL EXAMINATION

Mathematics Extension 2

Solutions

Section I

Answers

- | | | | |
|---|-----|----|-----|
| 1 | (B) | 6 | (C) |
| 2 | (B) | 7 | (B) |
| 3 | (B) | 8 | (A) |
| 4 | (B) | 9 | (A) |
| 5 | (C) | 10 | (D) |

Brief explanations

- 1 Equating the first and second components gives

$$\begin{aligned}1 + 2\lambda &= 2 + 4\mu \\ 2 + 4\lambda &= 3 + 8\mu\end{aligned}$$

Doubling the first equation and subtracting the second equation gives $0 = 1$ which is false, hence no solution exists and therefore no point of intersection exists.

Consider the direction vectors $\begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix}$. The latter can be written as $2 \begin{pmatrix} 2 \\ 4 \\ \frac{1}{2} \end{pmatrix}$ so the lines cannot be parallel. Hence, the answer is (B).

- 2 Let $z = x + iy$.

The first inequality gives $x \leq 0$. The second inequality gives $-\frac{1}{\sqrt{3}} \leq \frac{y-1}{x+1} \leq \frac{1}{\sqrt{3}}$.

This forms a triangle with base length of 1 and a perpendicular height of $\frac{2}{\sqrt{3}}$, giving an area of $\frac{1}{\sqrt{3}}$. Hence, the answer is (B).

- 3 Consider the counterexample $f(x) = x$ and $g(x) = \frac{\ln a}{\ln x}$ for some positive constant a . $(f(x))^{g(x)}$ simplifies to just some positive constant a . Therefore, neither (A) nor (C) are candidate answers.

Any real power of a non-negative number is non-negative, so the answer cannot be (D). Hence, the answer is (B).

- 4 As the stationary points are in the first and second quadrants, there is only one real root. Since $P(x)$ is monic then as $x \rightarrow -\infty$ then $P(x) \rightarrow -\infty$ and as $x \rightarrow \infty$ then $P(x) \rightarrow \infty$. However, at the stationary points $P(x) \geq 0$, which suggests that $P(x)$ has an x -intercept between the second and the third quadrant.

This means neither (A) nor (C) are correct. Since the remaining roots are non-real, they must occur in complex conjugates which is shown in (B).

- 5 The initial velocity is positive, as the object is projected upwards. This means neither (B) nor (D) are correct. Resistance and gravitational forces opposing the object's motion mean that the velocity decreases on the way up. It will eventually reach a maximum height at rest before falling back down. The conditions for terminal velocity do not occur until the object falls back down. Hence, the answer is (C).

- 6 Since $\underline{a}$ is on the line and $\underline{p}$ is outside the line, consider the projection of the vector $\underline{a} - \underline{p}$ in the direction on $\hat{n}$. This is the vector $[(\underline{a} - \underline{p}) \cdot \hat{n}] \hat{n}$. The shortest distance is the perpendicular distance from $\underline{p}$ to the line, hence the vector representing that is $[(\underline{a} - \underline{p}) \cdot \hat{n}] \hat{n} - (\underline{a} - \underline{p})$. The perpendicular distance is the length of that vector, hence the answer is (C).

- 7 The general form of the acceleration equation is $\ddot{x} = -n^2(x - c)$ where c is the centre of motion and the period is $\frac{2\pi}{n}$. Comparing this to the given equation $\ddot{x} = -kx$, it can be deduced that the centre is the origin and the period is $\frac{2\pi}{\sqrt{k}}$ which is a known value. Since the particle is initially at the origin, the initial acceleration can be deduced as zero. Consider a general solution to the acceleration equation is $x = a \sin(\sqrt{k}t + \alpha)$. When $t = 0$, $x = 0$ so $\alpha = 0$, which means that $x = a \sin(\sqrt{k}t)$. This equation still satisfies the acceleration equation and initial conditions for any value of a . Hence, the amplitude cannot be specifically determined, so the answer is (B).

- 8 Translating the first part of the logic notation, which is identical for all choices

For every positive ϵ there exists a positive δ , such that for every x in the domain $D...$

Choices (B) and (D) are not correct because of the $|f(x)| > \epsilon$ component. This suggests that $|f(x)|$ is larger than every positive number in the domain. This does not imply it is approaching zero as x approaches zero, but rather potentially approaching $\pm\infty$.

The remaining options are now based on deducing whether $|x| < \delta$ or $|x| > \delta$ is correct. Consider a simple example where $f(x) = x^2$ so this limit result is true. If (C) is correct then say for $\epsilon = 0.01$, we should be able to find a positive δ satisfying $|x| > \delta$ so that $x^2 < 0.01$. The latter inequality implies $-0.1 < x < 0.1$. However, it is not possible to find a positive δ which satisfies $|x| > \delta$ and gives $-0.1 < x < 0.1$.

Instead, it is possible to find a positive δ satisfying $|x| < \delta$ so that $x^2 < 0.01$ (namely $\delta = 0.1$). Hence, the answer is (A).

- 9 Use the fact that $z\bar{z} = |z|^2$ as well as $\bar{z}_1 + \bar{z}_2 = \overline{z_1 + z_2}$. After adding $a\bar{a}$ to both sides of the equality, each option can be rewritten as follows:

$$(A) \quad (a - b)(\overline{a - b}) = (a - c)(\overline{a - c}) \quad (B) \quad (a + b)(\overline{a + b}) = (a - c)(\overline{a - c})$$

$$(C) \quad (a - b)(\overline{a - b}) = (a + c)(\overline{a + c}) \quad (D) \quad (a + b)(\overline{a + b}) = (a + c)(\overline{a + c})$$

Using the fact that $\sqrt{z\bar{z}} = |z|$, these are equivalent to:

$$(A) \quad |a - b| = |a - c| \quad (B) \quad |a + b| = |a - c|$$

$$(C) \quad |a - b| = |a + c| \quad (D) \quad |a + b| = |a + c|$$

The only option that is related to the sides of the triangle formed by a, b and c is option (A) which implies that $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are equal in length, or equivalently $\triangle ABC$ is isosceles.

- 10 The correct answer is (D). Note that the following choices are actually equivalent.

$$(A^*) \quad \int_0^{\frac{\pi}{4}} \frac{2 \tan x}{\tan 2x} dx$$

$$(B^*) \quad \int_0^1 2x \tan^{-1} x dx$$

$$(C^*) \quad \int_0^{\frac{\pi}{4}} \frac{2x \sin x}{\cos^3 x} dx$$

$$(D) \quad \int_0^1 \left(\frac{1}{2} \ln(1 + x^2) + \tan^{-1} x \right) dx$$

(A*) follows from substituting $x = \tan \theta$ and changing dummy variables, then manipulating with the tangent double angle identity.

(B*) follows from integrating by parts, using $u = 1 - x^2$ and $dv = \frac{1}{1 + x^2} dx$.

(C*) follows from (B*) by substituting $x = \tan \theta$ and changing dummy variables.

(D) follows from using integration by parts using $u = 1 - x$ and $dv = \frac{1 + x}{1 + x^2} dx$.

$$\begin{aligned} \int_0^1 \frac{1 - x^2}{1 + x^2} dx &= \left[(1 - x) \left(\tan^{-1} x + \frac{1}{2} \ln(1 + x^2) \right) \right]_0^1 + \int_0^1 \left(\tan^{-1} x + \frac{1}{2} \ln(1 + x^2) \right) dx \\ &= \int_0^1 \left(\tan^{-1} x + \frac{1}{2} \ln(1 + x^2) \right) dx \end{aligned}$$

Question 11

- (a) Let $x = \frac{1}{u} \Rightarrow dx = -\frac{1}{u^2} du$. As $x \rightarrow 0^+$, $u \rightarrow \infty$ and as $x \rightarrow \infty$, $u \rightarrow 0$.

Applying this substitution and letting

$$\begin{aligned}
 I &= \int_0^\infty \frac{1}{x^8 + x^4 + 1} dx \\
 &= - \int_\infty^0 \frac{1}{\frac{1}{u^8} + \frac{1}{u^4} + 1} \times \frac{1}{u^2} du \\
 &= \int_0^\infty \frac{u^6}{u^8 + u^4 + 1} du \\
 &= \int_0^\infty \frac{x^6}{x^8 + x^4 + 1} dx \quad \text{since } u \text{ is a dummy variable} \\
 2I &= \int_0^\infty \frac{x^6}{x^8 + x^4 + 1} dx + \int_0^\infty \frac{1}{x^8 + x^4 + 1} dx \\
 I &= \frac{1}{2} \int_0^\infty \frac{x^6 + 1}{x^8 + x^4 + 1} dx
 \end{aligned}$$

The given identity can be used for $a = x^2$ to obtain $x^6 + 1 = (x^4 - x^2 + 1)(x^2 + 1)$. The denominator can also be manipulated as follows.

$$\begin{aligned}
 x^8 + x^4 + 1 &= (x^8 + 2x^4 + 1) - x^4 \\
 &= (x^4 + 1)^2 - (x^2)^2 \\
 &= (x^4 - x^2 + 1)(x^4 + x^2 + 1)
 \end{aligned}$$

This means that

$$\begin{aligned}
 I &= \frac{1}{2} \int_0^\infty \frac{(x^2 + 1)(x^4 - x^2 + 1)}{(x^4 - x^2 + 1)(x^4 + x^2 + 1)} dx \\
 &= \frac{1}{2} \int_0^\infty \frac{x^2 + 1}{x^4 + x^2 + 1} dx \\
 &= \frac{1}{2} \int_0^\infty \frac{1 + \frac{1}{x^2}}{x^2 + 1 + \frac{1}{x^2}} dx \\
 &= \frac{1}{2} \int_0^\infty \frac{1 + \frac{1}{x^2}}{(x - \frac{1}{x})^2 + 3} dx
 \end{aligned}$$

Consider the substitution $y = x - \frac{1}{x} \Rightarrow dy = \left(1 + \frac{1}{x^2}\right) dx$.

As $x \rightarrow 0^+$, $y \rightarrow -\infty$ and as $x \rightarrow \infty$, $y \rightarrow \infty$. Applying this substitution gives

$$\begin{aligned}
 I &= \frac{1}{2} \int_{-\infty}^\infty \frac{1}{y^2 + 3} dy \\
 &= \frac{1}{2\sqrt{3}} \left[\tan^{-1} \frac{y}{\sqrt{3}} \right]_{-\infty}^\infty \\
 &= \frac{\pi}{2\sqrt{3}}
 \end{aligned}$$

Remark: Another approach to evaluating $\int_0^\infty \frac{x^2 + 1}{x^4 + x^2 + 1} dx$ could be to notice that

$$\begin{aligned} x^4 + x^2 + 1 &= (x^4 + 2x^2 + 1) - x^2 \\ &= (x^2 + 1)^2 - x^2 \\ &= (x^2 - x + 1)(x^2 + x + 1) \end{aligned}$$

Using some algebraic manipulation

$$\begin{aligned} \int_0^\infty \frac{x^2 + 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \int_0^\infty \frac{2x^2 + 2 - x + x}{(x^2 - x + 1)(x^2 + x + 1)} dx \\ &= \frac{1}{2} \int_0^\infty \frac{x^2 + x + 1 + x^2 - x + 1}{(x^2 - x + 1)(x^2 + x + 1)} dx \\ &= \frac{1}{2} \int_0^\infty \left(\frac{1}{x^2 + x + 1} + \frac{1}{x^2 - x + 1} \right) dx \\ &= \frac{1}{2} \int_0^\infty \left(\frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} + \frac{1}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}} \right) dx \\ &= \frac{1}{2} \times \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{2x + 1}{\sqrt{3}} \right) + \tan^{-1} \left(\frac{2x - 1}{\sqrt{3}} \right) \right]_0^\infty \\ &= \frac{\pi}{\sqrt{3}} \end{aligned}$$

Hence $I = \frac{\pi}{2\sqrt{3}}$

- (b) Let A and B represent the complex numbers z_1 and z_2 respectively on the complex plane.

Since the coefficients are real then the non-real roots must occur in conjugate pairs on the complex plane. These roots are

$$x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$$

The circle of the smallest radius that passes through A and B would be one where AB is the diameter of the circle. This means that the centre of the circle is the midpoint of AB and the radius is half the length of AB . Hence, the equation of the circle is

$$\left(x + \frac{b}{2a}\right)^2 + y^2 = \frac{4ac - b^2}{4a^2}$$

- (c) If the lines ℓ_1 and ℓ_2 have a point of intersection then these are the solutions to

$$\begin{aligned} \underline{a}_1 + \lambda_1 \underline{b}_1 &= \underline{a}_2 + \lambda_2 \underline{b}_2 \\ (\underline{a}_1 + \lambda_1 \underline{b}_1) \cdot \underline{n} &= (\underline{a}_2 + \lambda_2 \underline{b}_2) \cdot \underline{n} \\ \underline{a}_1 \cdot \underline{n} + \lambda_1 \underline{b}_1 \cdot \underline{n} &= \underline{a}_2 \cdot \underline{n} + \lambda_2 \underline{b}_2 \cdot \underline{n} \quad \text{but } \underline{n} \text{ is perpendicular to both } \underline{b}_1 \text{ and } \underline{b}_2 \\ \underline{a}_1 \cdot \underline{n} &= \underline{a}_2 \cdot \underline{n} \\ (\underline{a}_1 - \underline{a}_2) \cdot \underline{n} &= 0 \end{aligned}$$

- (d) (i) Starting from the acceleration equation

$$\begin{aligned}
 \ddot{x} &= -kv^n \\
 v \frac{dv}{dx} &= -kv^n \\
 \int \frac{dv}{v^{n-1}} &= -k \int dx \\
 -\frac{1}{(n-2)v^{n-2}} &= -kx + c \quad \text{when } x=0, v=u \Rightarrow c = -\frac{1}{(n-2)u^{n-2}} \\
 kx &= \frac{1}{(n-2)v^{n-2}} - \frac{1}{(n-2)u^{n-2}} \\
 x &= \frac{u^{n-2} - v^{n-2}}{k(n-2)(uv)^{n-2}}
 \end{aligned}$$

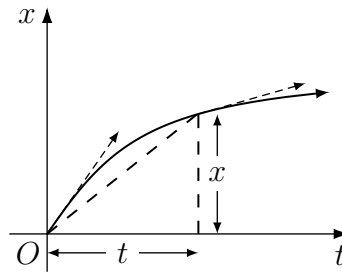
- (ii) Applying the same approach similarly for the time derivative of acceleration

$$\begin{aligned}
 \frac{dv}{dt} &= -kv^n \\
 t &= \frac{u^{n-1} - v^{n-1}}{k(n-1)(uv)^{n-1}}
 \end{aligned}$$

The average speed of the object from the origin to displacement x is

$$\begin{aligned}
 v_{ave} &= \frac{x-0}{t-0} \\
 &= \frac{u^{n-2} - v^{n-2}}{k(n-2)(uv)^{n-2}} \times \frac{k(n-1)(uv)^{n-1}}{u^{n-1} - v^{n-1}} \\
 &= \frac{a_{n-2}(uv)}{a_{n-1}}
 \end{aligned}$$

Note that the acceleration equation is always negative. This means that from the initial position, the object is slowing down while moving in the positive direction. This also means that a displacement-time graph of the object's motion will be concave down and increasing.



This means that the average speed (the gradient of the secant) at a given point is lower than the initial speed (slope of the tangent to the displacement curve at $t=0$) but higher than the instantaneous speed (slope of the tangent to the displacement curve at time $t>0$).

Hence

$$\begin{aligned}
v &< v_{ave} < u \\
v &< \frac{a_{n-2}(uv)}{a_{n-1}} < u \\
\frac{1}{u} &< \frac{a_{n-2}}{a_{n-1}} < \frac{1}{v} \\
v &< \frac{a_{n-1}}{a_{n-2}} < u
\end{aligned}$$

- (e) (i) By the method for partial fractions, there exist pairs of real constants $(b_1, c_1), (b_2, c_2), \dots, (b_{n-1}, c_{n-1})$ and (b_n, c_n) , such that

$$\frac{1}{P(x)} = \frac{b_1x + c_1}{x^2 + a_1^2} + \frac{b_2x + c_2}{x^2 + a_2^2} + \dots + \frac{b_nx + c_n}{x^2 + a_n^2}.$$

Notice that

$$\frac{1}{P(-x)} = \frac{-b_1x + c_1}{x^2 + a_1^2} + \frac{-b_2x + c_2}{x^2 + a_2^2} + \dots + \frac{-b_nx + c_n}{x^2 + a_n^2}.$$

But $P(x)$ is an even function so $P(-x) = P(x)$, hence

$$\begin{aligned}
\frac{1}{P(x)} + \frac{1}{P(-x)} &= \frac{2c_1}{x^2 + a_1^2} + \frac{2c_2}{x^2 + a_2^2} + \dots + \frac{2c_n}{x^2 + a_n^2} \\
\frac{1}{P(x)} &= \frac{c_1}{x^2 + a_1^2} + \frac{c_2}{x^2 + a_2^2} + \dots + \frac{c_n}{x^2 + a_n^2}
\end{aligned}$$

- (ii) Making c_1 the subject from part (i)

$$c_1 = \frac{1}{(x^2 + a_2^2)(x^2 + a_3^2) \dots (x^2 + a_n^2)} - (x^2 + a_1^2) \left(\frac{c_2}{x^2 + a_2^2} + \frac{c_3}{x^2 + a_3^2} + \dots + \frac{c_n}{x^2 + a_n^2} \right)$$

Let $x = ia_1$, which implies $x^2 + a_1^2 = 0$ hence

$$c_1 = \frac{1}{(a_2^2 - a_1^2)(a_3^2 - a_1^2) \dots (a_n^2 - a_1^2)}$$

Applying this approach similarly for all the other constants gives

$$\begin{aligned}
c_2 &= \frac{1}{(a_1^2 - a_2^2)(a_3^2 - a_2^2) \dots (a_n^2 - a_2^2)} \\
&\vdots \\
c_n &= \frac{1}{(a_1^2 - a_n^2)(a_2^2 - a_n^2) \dots (a_{n-1}^2 - a_n^2)}
\end{aligned}$$

Question 12

- (a) Let the principal root be $z_1 = e^{\frac{2\pi i}{n}}$, and note that $z_1^n = 1$ and $z_k = z_1^k$

$$\begin{aligned}
 & (z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_4)^2 + \dots + (z_{n-1} - z_n)^2 + (z_n - z_1)^2 \\
 &= (z_1 - z_1^2)^2 + (z_1^2 - z_1^3)^2 + (z_1^3 - z_1^4)^2 + \dots + (z_1^{n-1} - z_1^n)^2 + (z_1^n - z_1)^2 \\
 &= z_1^2(1 - z_1)^2 + z_1^4(1 - z_1)^2 + z_1^6(1 - z_1)^2 + \dots + z_1^{2(n-1)}(1 - z_1)^2 + (1 - z_1)^2 \\
 &= (1 - z_1)^2(1 + z_1^2 + z_1^4 + \dots + z_1^{2(n-1)}) \\
 &= (1 - z_1)^2 \times \frac{z_1^{2n} - 1}{z_1^2 - 1} \\
 &= 0 \quad \text{since } z_1^n = 1 \Rightarrow z_1^{2n} = 1
 \end{aligned}$$

- (b) (i) Notice that each component of the acceleration shows simple harmonic motion about the origin ($a_x = -n^2x$ and $a_y = -n^2y$). Hence, the general solutions for the displacement equations can be used.

$$\begin{aligned}
 x &= \hat{a} \cos(nt + \alpha) \\
 v_x &= -\hat{a}n \sin(nt + \alpha) \\
 \text{when } t = 0, x &= A, v_x = 0 \\
 A &= \hat{a} \cos \alpha \\
 0 &= -\hat{a}n \sin \alpha
 \end{aligned}$$

Since $\hat{a} \neq 0$ and $n \neq 0$ then $\alpha = 0$, which means that $\hat{a} = A$.
Similarly, consider the general vertical displacement equation

$$\begin{aligned}
 y &= \hat{b} \cos(nt + \beta) \\
 v_y &= -\hat{b}n \sin(nt + \beta) \\
 \text{when } t = 0, y &= B \cos \theta, v_y = -Bn \sin \theta \\
 B \cos \theta &= \hat{b} \cos \beta \\
 B \sin \theta &= \hat{b} \sin \beta \\
 B^2(\cos^2 \theta + \sin^2 \theta) &= \hat{b}^2(\cos^2 \beta + \sin^2 \beta) \\
 \Rightarrow \hat{b} &= B \\
 \tan \theta &= \tan \beta \\
 \Rightarrow \beta &= \theta
 \end{aligned}$$

Hence the displacement vector is

$$\underline{r} = A \cos(nt) \underline{i} + B \cos(nt + \theta) \underline{j}$$

Remark: Could have chosen $\hat{b} = -B$ instead, which would lead to a different value for β but lead to an equivalent equation.

- (ii) From the displacement equations $x = A \cos(nt)$ and $y = A \cos(nt + \theta)$

$$\begin{aligned}
LHS &= \frac{x^2}{A^2} - \frac{2xy \cos \theta}{AB} + \frac{y^2}{B^2} \\
&= \frac{A^2 \cos^2(nt)}{A^2} - \frac{2AB \cos(nt) \cos(nt + \theta) \cos \theta}{AB} + \frac{B^2 \cos^2(nt + \theta)}{B^2} \\
&= \cos^2(nt) + \cos(nt + \theta)[\cos(nt + \theta) - 2 \cos(nt) \cos \theta] \\
&= \cos^2(nt) + \cos(nt + \theta)[\cos(nt) \cos \theta - \sin(nt) \sin \theta - 2 \cos(nt) \cos \theta] \\
&= \cos^2(nt) - \cos(nt + \theta)[\cos(nt) \cos \theta + \sin(nt) \sin \theta] \\
&= \cos^2(nt) - \cos(nt + \theta) \cos(nt - \theta) \\
&= \cos^2(nt) - [\cos(nt) \cos \theta - \sin(nt) \sin \theta][\cos(nt) \cos \theta + \sin(nt) \sin \theta] \\
&= \cos^2(nt) - \cos^2(nt) \cos^2 \theta + \sin^2(nt) \sin^2 \theta \\
&= \cos^2(nt)(1 - \cos^2 \theta) + \sin^2(nt) \sin^2 \theta \\
&= \sin^2 \theta (\cos^2(nt) + \sin^2(nt)) \\
&= \sin^2 \theta \\
&= RHS
\end{aligned}$$

Hence, the Cartesian equation of the particle's path is

$$\frac{x^2}{A^2} - \frac{2xy \cos \theta}{AB} + \frac{y^2}{B^2} = \sin^2 \theta.$$

- (iii) When $\theta = \frac{\pi}{2}$, the particle's path follows the following Cartesian equation

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$$

This curve has x -intercepts $(-A, 0)$ and $(A, 0)$, and y -intercepts $(0, B)$ and $(0, -B)$ (it is in fact an ellipse, which reduces to a circle if $A = B$).

From the conditions given in the question, the particle is initially at $(A, 0)$ with velocity $\underline{v} = -Bn\hat{j}$. This means that the particle starts at the positive x -intercept and is moving in the negative y direction. As it continues to move along the x - y plane, it will trace the path in a clockwise direction.

- (c) Since the $P(x)$ has real coefficients, it can either have three real roots or two complex conjugate roots and one real root. However, there are only two real roots which satisfy $|x^3| = 8$ (namely $x = \pm 2$). This means that $P(x)$ must have two complex conjugate roots and one real root.

Given that $\alpha + \beta = 2$, the roots α and β must be the two complex conjugate roots, in order to sum to a real number. Let these be $1 \pm ia$ for some real a .

$$\begin{aligned}
|(1 + ia)^3| &= 8 \\
|1 + ia|^3 &= 8 \\
(\sqrt{1 + a^2})^3 &= 8 \\
1 + a^2 &= 4 \quad \text{noting that } a \text{ is real} \\
a &= \pm\sqrt{3}
\end{aligned}$$

This means that the two complex conjugate roots of $P(x)$ are $1 \pm i\sqrt{3}$. Let the real root be γ . Taking the sum of the roots in pairs

$$\begin{aligned}\alpha\beta + \beta\gamma + \alpha\gamma &= c \\ (1 + i\sqrt{3})(1 - i\sqrt{3}) + \gamma(\alpha + \beta) &= c \\ 4 + 2\gamma &= c\end{aligned}$$

The only real roots to $|x^3| = 8$ are $x = \pm 2$. However, $\gamma \neq -2$ since c must be a non-zero coefficient. Hence, $\gamma = 2$ is the real root. This gives $c = 8$. To find the remaining coefficients b and d , use sum and product of roots.

$$\begin{aligned}\alpha + \beta + \gamma &= -b \Rightarrow b = -4 \\ \alpha\beta\gamma &= -d \Rightarrow d = -8\end{aligned}$$

Hence $b = -4, c = 8$ and $d = -8$.

(d) (i) If $\underline{u} \cdot \underline{v} = 0$ then consider

$$\begin{aligned}(a_1\underline{u} + a_2\underline{v}) \cdot \underline{v} &= 0 \cdot \underline{v} \\ a_1\underline{u} \cdot \underline{v} + a_2\underline{v} \cdot \underline{v} &= 0 \\ a_2|\underline{v}|^2 &= 0\end{aligned}$$

Since $\underline{v} \neq 0$ then $|\underline{v}|^2 > 0$ so it must be that $a_2 = 0$. This, in turn, implies that

$$a_1\underline{u} = 0$$

Since $\underline{u} \neq 0$ then it must be that $a_1 = 0$. Hence, the statement is always true.

(ii) The converse statement is

If the only solution to $a_1\underline{u} + a_2\underline{v} = 0$ is $a_1 = a_2 = 0$ then $\underline{u} \cdot \underline{v} = 0$

Consider

$$\underline{u} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \underline{v} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

The vector equation $a_1\underline{u} + a_2\underline{v} = 0$ becomes

$$\begin{pmatrix} a_1 + 2a_2 \\ a_1 + a_2 \\ a_1 + 2a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The **only** way this is possible is if $a_1 = a_2 = 0$ (i.e. there are no other possible values of a_1 and a_2 which could satisfy the equations). However, $\underline{u} \cdot \underline{v} = 5$ which is non-zero. This is a counterexample to the converse statement. Hence, the converse statement is not necessarily true.

Question 13

- (a) (i) Rearranging the given relation

$$\begin{aligned} z_3 &= \frac{az_1 + ibz_2}{a + ib} \\ az_3 + ibz_3 &= az_1 + ibz_2 \\ -a(z_1 - z_3) &= ib(z_2 - z_3) \\ z_1 - z_3 &= -\frac{ib}{a}(z_2 - z_3) \end{aligned}$$

Since the vectors $\overrightarrow{CB}$ (represented by $z_2 - z_3$) and $\overrightarrow{CA}$ (represented by $z_1 - z_3$) are related by multiplication of i and a scalar, then CB and CA are right angles relative to each other. Hence, $\triangle ACB$ is a right-angled triangle.

- (ii) For $\triangle ACB$ to be isosceles $CA = CB$ is required so

$$\begin{aligned} |z_1 - z_3| &= |z_2 - z_3| \\ \left| -\frac{ib}{a}(z_2 - z_3) \right| &= |z_2 - z_3| \\ |a| &= |b| \end{aligned}$$

Hence a and b must be equal in magnitude for $\triangle ACB$ to be isosceles.

- (b) (i) Using the given inequality, let $x = ab, y = bc$ and $z = ca$

$$\begin{aligned} \frac{ab + bc + ca}{3} &\geq (abc)^{\frac{2}{3}} \quad \text{but } ab + bc + ca = 3 \\ (abc)^{\frac{2}{3}} &\leq 1 \\ abc &\leq 1 \end{aligned}$$

- (ii) Since $abc \leq 1$ from part (i) and $ab + bc + ca = 3$

$$\begin{aligned} \frac{1}{1 + a^2(b + c)} &= \frac{1}{1 + a(ab + ac)} \\ &= \frac{1}{1 + a(3 - bc)} \\ &= \frac{1}{1 - abc + 3a} \\ &\leq \frac{1}{3a} \end{aligned}$$

Similarly applying this to the other fractions gives

$$\begin{aligned} \frac{1}{1 + a^2(b + c)} + \frac{1}{1 + b^2(a + c)} + \frac{1}{1 + c^2(a + b)} &\leq \frac{1}{3a} + \frac{1}{3b} + \frac{1}{3c} \\ &= \frac{ab + bc + ca}{3abc} \\ &= \frac{1}{abc} \end{aligned}$$

(c) (i) Using the definition of $H_1(x)$ for $n = 1$

$$\begin{aligned}
LHS &= H_1(x) \\
&= (-1)^1 e^{x^2} \frac{d}{dx} (e^{-x^2}) \\
&= -e^{x^2} (-2xe^{-x^2}) \\
&= 2x \\
RHS &= 2xH_0(x) - \frac{d}{dx}H_0(x) \quad \text{but } H_0(x) = 1 \\
&= 2x
\end{aligned}$$

$LHS = RHS$ so the statement is true for $n = 1$.

Assume the statement is true for $n = k$, that is

$$H_k(x) = 2xH_{k-1}(x) - \frac{d}{dx}H_{k-1}(x).$$

Required to prove that for $n = k + 1$

$$H_{k+1}(x) = 2xH_k(x) - \frac{d}{dx}H_k(x).$$

$$\begin{aligned}
LHS &= H_{k+1}(x) \\
&= (-1)^{k+1} e^{x^2} \frac{d^{k+1}}{dx^{k+1}} (e^{-x^2}) \\
&= (-1)^{k+1} e^{x^2} \frac{d}{dx} \left[\frac{d^k}{dx^k} (e^{-x^2}) \right] \\
&= (-1)^{k+1} e^{x^2} \frac{d}{dx} \left[\frac{H_k(x)}{(-1)^k e^{x^2}} \right] \\
&= -e^{x^2} \left[e^{-x^2} \frac{d}{dx} H_k(x) - 2xe^{-x^2} H_k(x) \right] \\
&= 2xH_k(x) - \frac{d}{dx}H_k(x) \\
&= RHS
\end{aligned}$$

Hence, the statement is true by induction for all positive integers n .

- (ii) If $P(x)$ and $Q(x)$ are some polynomials, then $P(x) + Q'(x)$ is also a polynomial (since $Q'(x)$ is a polynomial). From part (i), $H_0(x)$ and $H_1(x)$ are polynomials, hence $H_n(x)$ must be a polynomial.

The degree of $2xH_{n-1}(x)$ is one more than the degree of $H_{n-1}(x)$ and the degree of $\frac{d}{dx}H_{n-1}(x)$ is at least one less than the degree of $H_{n-1}(x)$.

Therefore, the degree of $H_n(x) = 2xH_{n-1}(x) - \frac{d}{dx}H_{n-1}(x)$ must be one more than the degree of $H_{n-1}(x)$ since the power of the leading term of $2xH_{n-1}(x)$ will never coincide (or cancel out) with a term from $\frac{d}{dx}H_{n-1}(x)$. In other words, $\deg(H_n(x)) = \deg(H_{n-1}(x)) + 1$. Since $H_0(x)$ is of degree zero and $H_1(x)$ is degree one, then $H_n(x)$ must be degree n when applying this recurrence relationship.

- (d) (i) If $\underline{x}$ is some variable point on the plane and $\underline{x}_0$ also lies on this plane, then the vector $\underline{x} - \underline{x}_0$ is parallel to the plane. This is perpendicular to the vector CP , which is represented by $\underline{x}_0 - \underline{c}$. Hence, their dot product must be zero.

$$(\underline{x} - \underline{x}_0) \cdot (\underline{x}_0 - \underline{c}) = 0$$

- (ii) From the equation of the sphere

$$\begin{aligned} |\underline{x} - \underline{c}| &= r \\ (\underline{x} - \underline{c}) \cdot (\underline{x} - \underline{c}) &= r^2 \\ \underline{x} \cdot \underline{x} - 2\underline{x} \cdot \underline{c} + \underline{c} \cdot \underline{c} &= r^2 \end{aligned}$$

From the equation of the plane

$$\begin{aligned} (\underline{x} - \underline{x}_0) \cdot (\underline{x}_0 - \underline{c}) &= 0 \\ \underline{x} \cdot \underline{x}_0 - \underline{x} \cdot \underline{c} - \underline{x}_0 \cdot \underline{x}_0 + \underline{x}_0 \cdot \underline{c} &= 0 \\ 2\underline{x} \cdot \underline{x}_0 - 2\underline{x} \cdot \underline{c} - 2\underline{x}_0 \cdot \underline{x}_0 + 2\underline{x}_0 \cdot \underline{c} &= 0 \end{aligned}$$

Subtracting the two equations

$$\begin{aligned} \underline{x} \cdot \underline{x} + \underline{c} \cdot \underline{c} - 2\underline{x} \cdot \underline{x}_0 + 2\underline{x}_0 \cdot \underline{x}_0 - 2\underline{x}_0 \cdot \underline{c} &= r^2 \\ \underline{x} \cdot \underline{x} - 2\underline{x} \cdot \underline{x}_0 + \underline{x}_0 \cdot \underline{x}_0 + \underline{x}_0 \cdot \underline{x}_0 - 2\underline{x}_0 \cdot \underline{c} + \underline{c} \cdot \underline{c} &= r^2 \\ (\underline{x} - \underline{x}_0) \cdot (\underline{x} - \underline{x}_0) + (\underline{x}_0 - \underline{c}) \cdot (\underline{x}_0 - \underline{c}) &= r^2 \\ |\underline{x} - \underline{x}_0|^2 + |\underline{x}_0 - \underline{c}|^2 &= r^2 \end{aligned}$$

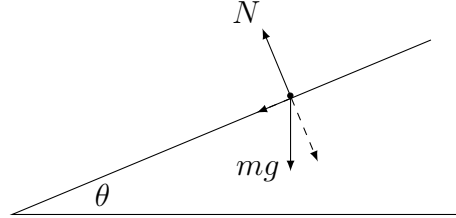
Since $\underline{x}_0$ lies on the sphere then $|\underline{x}_0 - \underline{c}|^2 = r^2$, hence

$$|\underline{x} - \underline{x}_0|^2 = 0$$

The solution(s) to $\underline{x}$ when solving the equations of the sphere and plane simultaneously should be the point(s) of intersection. From the above result there is only one solution for $\underline{x}$, namely $\underline{x}_0$. This means that the plane only touches the sphere at one point P (i.e. it is a tangent plane).

Question 14

- (a) (i) Projecting the gravitational force vector in the direction of the normal force gives a vector of magnitude $mg \cos \theta$.



The direction of the net force on the object is parallel to the inclined surface. This means that the net force perpendicular to the inclined surface must be zero. Hence, $N = mg \cos \theta$.

- (ii) Let x, v and a be the displacement, velocity and acceleration of the object respectively relative to its initial point. Define the direction of motion as the positive direction.

$$\begin{aligned} ma &= mg \sin \theta \\ \frac{d}{dx} \left(\frac{v^2}{2} \right) &= g \sin \theta \\ \frac{v^2}{2} &= gx \sin \theta + c \quad \text{when } x = 0, v = 0 \Rightarrow c = 0 \\ v^2 &= 2gx \sin \theta \end{aligned}$$

When $x = d$, the speed of the object at the base of the inclined surface is

$$v_d = \sqrt{2gd \sin \theta}.$$

- (iii) From the acceleration equation

$$\begin{aligned} \frac{dv}{dt} &= g \sin \theta \\ v &= gt \sin \theta + c \quad \text{when } t = 0, v = 0 \Rightarrow c = 0 \\ v &= gt \sin \theta \end{aligned}$$

When $t = t_d$ and $v = v_d$ so

$$\begin{aligned} v_d &= gt_d \sin \theta \\ v_d^2 &= gv_d t_d \sin \theta \quad \text{but } v_d = \sqrt{2gd \sin \theta} \\ 2gd \sin \theta &= gv_d t_d \sin \theta \\ d &= \frac{v_d t_d}{2} \end{aligned}$$

- (b) (i) From the geometric series and the results $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$

$$\begin{aligned}
\sum_{k=0}^n z^k &= \frac{z^{n+1} - 1}{z - 1} \times \frac{z^{-\frac{n+1}{2}}}{z^{-\frac{n+1}{2}}} \\
&= \frac{z^{\frac{n+1}{2}} - z^{-\frac{n+1}{2}}}{z^{-\frac{n}{2}}(z^{\frac{1}{2}} - z^{-\frac{1}{2}})} \\
&= \left(\frac{\cos\left(\frac{(n+1)\theta}{2}\right) + i \sin\left(\frac{(n+1)\theta}{2}\right) - \cos\left(-\frac{(n+1)\theta}{2}\right) - i \sin\left(-\frac{(n+1)\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right) + i \sin\left(\frac{\theta}{2}\right) - \cos\left(-\frac{\theta}{2}\right) - i \sin\left(-\frac{\theta}{2}\right)} \right) z^{\frac{n}{2}} \\
&= \left(\frac{2i \sin\left(\frac{(n+1)\theta}{2}\right)}{2i \sin\frac{\theta}{2}} \right) z^{\frac{n}{2}} \\
\sum_{k=0}^n \operatorname{Re}(z^k) &= \left(\frac{\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\frac{\theta}{2}} \right) \operatorname{Re}(z^{\frac{n}{2}})
\end{aligned}$$

Noting that $\operatorname{Re}\left(\sum_{k=0}^n z^k\right) = \sum_{k=0}^n \operatorname{Re}(z^k)$ when equating the real parts.

- (ii) Evaluating the real parts in terms of real trigonometric functions and converting trigonometric products to sums

$$\begin{aligned}
\sum_{k=0}^n \operatorname{Re}(z^k) &= \left(\frac{\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\frac{\theta}{2}} \right) \operatorname{Re}(z^{\frac{n}{2}}) \\
\sum_{k=0}^n \cos(k\theta) &= \left(\frac{\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\frac{\theta}{2}} \right) \cos\frac{n\theta}{2} \\
&= \frac{\sin\left(\frac{(2n+1)\theta}{2}\right) + \sin\frac{\theta}{2}}{2 \sin\frac{\theta}{2}} \\
&= \frac{\sin\left(\frac{(2n+1)\theta}{2}\right)}{2 \sin\frac{\theta}{2}} + \frac{1}{2}
\end{aligned}$$

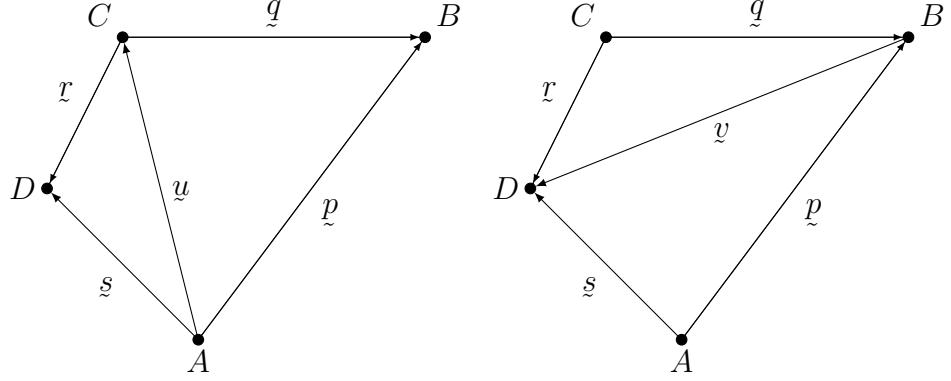
Differentiate both sides with respect to θ

$$\begin{aligned}
-\sum_{k=1}^n k \sin(k\theta) &= \frac{(2n+1) \cos\left(\frac{(2n+1)\theta}{2}\right) \sin\frac{\theta}{2} - \sin\left(\frac{(2n+1)\theta}{2}\right) \cos\frac{\theta}{2}}{4 \sin^2\frac{\theta}{2}} \\
\sum_{k=1}^n k \sin(k\theta) &= \frac{\left[\sin\left(\frac{(2n+1)\theta}{2}\right) \cos\frac{\theta}{2} - \cos\left(\frac{(2n+1)\theta}{2}\right) \sin\frac{\theta}{2} \right] - 2n \cos\left(\frac{(2n+1)\theta}{2}\right) \sin\frac{\theta}{2}}{2 \left(2 \sin^2\frac{\theta}{2}\right)} \\
&= \frac{\sin(n\theta) - n(\sin[(n+1)\theta] - \sin(n\theta))}{2(1 - \cos\theta)} \\
&= \frac{(n+1) \sin(n\theta) - n \sin[(n+1)\theta]}{2(1 - \cos\theta)}
\end{aligned}$$

- (c) (i) Since $\angle ABC + \angle ADC = \pi$ then $\angle BCD + \angle BAD = \pi$, by angle sum of the quadrilateral $ABCD$.

For ease of notation, denote the following vectors as follows

$$\overrightarrow{AB} = \underline{p} \quad \overrightarrow{CB} = \underline{q} \quad \overrightarrow{CD} = \underline{r} \quad \overrightarrow{AD} = \underline{s} \quad \overrightarrow{AC} = \underline{u} \quad \overrightarrow{BD} = \underline{v}$$



From the cosine rule (or using the fact that $\underline{u} = \underline{q} - \underline{p} \Rightarrow |\underline{u}|^2 = (\underline{q} - \underline{p}) \cdot (\underline{q} - \underline{p})$)

$$\cos \angle ABC = \frac{|\underline{p}|^2 + |\underline{q}|^2 - |\underline{u}|^2}{2|\underline{p}||\underline{q}|}$$

$$\text{Similarly} \quad \cos \angle ADC = \frac{|\underline{r}|^2 + |\underline{s}|^2 - |\underline{u}|^2}{2|\underline{r}||\underline{s}|}$$

Since $\angle ABC + \angle ADC = \pi$ and $\cos(\pi - x) = -\cos x$ then

$$\begin{aligned} \cos \angle ABC + \cos \angle ADC &= \cos \angle ABC + \cos(\pi - \angle ABC) \\ &= 0 \end{aligned}$$

Hence

$$\begin{aligned} \frac{|\underline{p}|^2 + |\underline{q}|^2 - |\underline{u}|^2}{|\underline{p}||\underline{q}|} + \frac{|\underline{r}|^2 + |\underline{s}|^2 - |\underline{u}|^2}{|\underline{r}||\underline{s}|} &= 0 \\ (|\underline{p}|^2 + |\underline{q}|^2)|\underline{r}||\underline{s}| + (|\underline{r}|^2 + |\underline{s}|^2)|\underline{p}||\underline{q}| &= |\underline{u}|^2(|\underline{p}||\underline{q}| + |\underline{r}||\underline{s}|) \end{aligned}$$

Expanding and manipulating gives

$$\begin{aligned} |\underline{u}|^2 &= \frac{|\underline{p}||\underline{s}|(|\underline{p}||\underline{r}| + |\underline{q}||\underline{s}|) + |\underline{q}||\underline{r}|(|\underline{p}||\underline{r}| + |\underline{q}||\underline{s}|)}{|\underline{p}||\underline{q}| + |\underline{r}||\underline{s}|} \\ &= \frac{(|\underline{p}||\underline{s}| + |\underline{q}||\underline{r}|)(|\underline{p}||\underline{r}| + |\underline{q}||\underline{s}|)}{|\underline{p}||\underline{q}| + |\underline{r}||\underline{s}|} \end{aligned}$$

Similarly applying the above derivation to the other diagonal gives

$$|\underline{v}|^2 = \frac{(|\underline{q}||\underline{p}| + |\underline{r}||\underline{s}|)(|\underline{q}||\underline{s}| + |\underline{r}||\underline{p}|)}{|\underline{q}||\underline{r}| + |\underline{s}||\underline{p}|}$$

Hence

$$\begin{aligned} |\underline{u}|^2 |\underline{v}|^2 &= \frac{(|\underline{p}||\underline{s}| + |\underline{q}||\underline{r}|)(|\underline{p}||\underline{r}| + |\underline{q}||\underline{s}|)}{|\underline{p}||\underline{q}| + |\underline{r}||\underline{s}|} \times \frac{(|\underline{p}||\underline{q}| + |\underline{r}||\underline{s}|)(|\underline{q}||\underline{s}| + |\underline{r}||\underline{p}|)}{|\underline{q}||\underline{r}| + |\underline{s}||\underline{p}|} \\ &= (|\underline{p}||\underline{r}| + |\underline{q}||\underline{s}|)^2 \end{aligned}$$

Noting that $|p||r| + |q||s|, |u|$ and $|v|$ are all positive then

$$|p||r| + |q||s| = |u||v|$$

From the position vector definitions

$$\overrightarrow{AB} = \underline{b} - \underline{a} \quad \overrightarrow{CB} = \underline{b} - \underline{c} \quad \overrightarrow{CD} = \underline{d} - \underline{c} \quad \overrightarrow{AD} = \underline{d} - \underline{a} \quad \overrightarrow{AC} = \underline{c} - \underline{a} \quad \overrightarrow{BD} = \underline{d} - \underline{b}$$

Hence

$$|\underline{b} - \underline{a}||\underline{d} - \underline{c}| + |\underline{d} - \underline{a}||\underline{c} - \underline{b}| = |\underline{c} - \underline{a}||\underline{d} - \underline{b}|.$$

- (ii) Since three distinct non-collinear points will always uniquely define a plane, consider the case (without loss of generality) where A, B and C form a plane and D does not lie on the plane.

There exists a point D^* (represented by position vector $\underline{d}^*$) on the plane of A, B and C such that $AD = AD^*$ and $CD = CD^*$ and $\angle ADC = \angle AD^*C$. This forms a quadrilateral $ABCD^*$ and the following holds as shown in part (i).

$$|\underline{b} - \underline{a}||\underline{d}^* - \underline{c}| + |\underline{d}^* - \underline{a}||\underline{c} - \underline{b}| = |\underline{c} - \underline{a}||\underline{d}^* - \underline{b}|$$

But $AD = AD^*$ and $CD = CD^*$ so $|\underline{d}^* - \underline{c}| = |\underline{d} - \underline{c}|$ and $|\underline{d}^* - \underline{a}| = |\underline{d} - \underline{a}|$

$$|\underline{b} - \underline{a}||\underline{d} - \underline{c}| + |\underline{d} - \underline{a}||\underline{c} - \underline{b}| = |\underline{c} - \underline{a}||\underline{d}^* - \underline{b}|$$

However, $BD^* > BD$ (see remark) which means that $|\underline{d}^* - \underline{b}| > |\underline{d} - \underline{b}|$ so

$$|\underline{b} - \underline{a}||\underline{d} - \underline{c}| + |\underline{d} - \underline{a}||\underline{c} - \underline{b}| > |\underline{c} - \underline{a}||\underline{d} - \underline{b}|$$

Remark: An intuitive way to visualise this is to consider a flat rectangular piece of paper labelled $ABCD$. If one of the vertices of the paper, say D , is lifted up but the other vertices stay where they are, then notice that the direct distance from vertex B to the vertex D has shortened, compared to the original positioning on the plane of the paper.

Question 15

- (a) (i) Using integration by parts for $n \geq 2$

$$\begin{aligned}
 I_n &= \int_0^{\frac{\pi}{2}} \sin^n x \, dx \\
 &= [-\sin^{n-1} x \cos x]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx \\
 &= (n-1)(I_{n-2} - I_n) \\
 I_n &= \frac{n-1}{n} I_{n-2}
 \end{aligned}$$

- (ii) From part (i)

$$\begin{aligned}
 I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\
 &= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} I_{2n-4} \\
 &\vdots \\
 &= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \cdots \times \frac{3}{4} I_2 \\
 &= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \cdots \times \frac{3}{4} \times \frac{1}{2} I_0 \\
 &= \frac{(2n-1) \times (2n-3) \times \cdots \times 3 \times 1}{2^n n!} \int_0^{\frac{\pi}{2}} dx \times \frac{2n \times (2n-2) \times \cdots \times 4 \times 2}{2n \times (2n-2) \times \cdots \times 4 \times 2} \\
 &= \frac{\pi}{2} \times \frac{(2n)!}{2^{2n} (n!)^2}
 \end{aligned}$$

- (iii) For the domain $[0, \frac{\pi}{2}]$ note that

$$\begin{aligned}
 0 &\leq \sin x \leq 1 \\
 \sin^{2n+1} x &\leq \sin^{2n} x \\
 \sin^{2n+2} x &\leq \sin^{2n+1} x \\
 \Rightarrow \sin^{2n+2} x &\leq \sin^{2n+1} x \leq \sin^{2n} x \\
 \int_0^{\frac{\pi}{2}} \sin^{2n+2} x \, dx &\leq \int_0^{\frac{\pi}{2}} \sin^{2n+1} x \, dx \leq \int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx \\
 I_{2n+2} &\leq I_{2n+1} \leq I_{2n} \\
 \frac{2n+1}{2n+2} I_{2n} &\leq I_{2n+1} \leq I_{2n} \\
 \frac{2n+1}{2n+2} &\leq \frac{I_{2n+1}}{I_{2n}} \leq 1
 \end{aligned}$$

Noting that since $0 \leq \sin x \leq 1$ in the domain $[0, \frac{\pi}{2}]$ then $I_{2n} > 0$.

(iv) Consider the ratio

$$\begin{aligned}\frac{I_{2n+1}}{I_{2n}} &= \frac{2^{2n}(n!)^2}{(2n+1)!} \times \frac{2}{\pi} \times \frac{2^{2n}(n!)^2}{(2n)!} \\ &= \frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} \times \frac{2}{\pi}\end{aligned}$$

From part (iii), when $n \rightarrow \infty$ then $\frac{2n+1}{2n+2} \rightarrow 1$ so $\frac{I_{2n+1}}{I_{2n}} \rightarrow 1$, hence

$$\lim_{n \rightarrow \infty} \frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} = \frac{\pi}{2}$$

(v) Consider the ratio

$$\begin{aligned}\frac{(u_n)^4}{(u_{2n})^2} &= \frac{e^{4n}(n!)^4}{n^{4n+2}} \times \frac{(2n)^{4n+1}}{e^{4n}[(2n)!]^2} \\ &= \frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} \times \frac{2(2n+1)}{n} \\ &= \frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} \times \left(4 + \frac{2}{n}\right)\end{aligned}$$

As $n \rightarrow \infty$ then $4 + \frac{2}{n} \rightarrow 4$ and $\frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} \rightarrow \frac{\pi}{2}$ from part (iv). Hence

$$\lim_{n \rightarrow \infty} \frac{(u_n)^4}{(u_{2n})^2} = 2\pi$$

However, when $n \rightarrow \infty$ then $u_n \rightarrow L$ and similarly when $2n \rightarrow \infty$ then $u_{2n} \rightarrow L$. This means that

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{(u_n)^4}{(u_{2n})^2} &= \frac{L^4}{L^2} \\ L^2 &= 2\pi \\ L &= \sqrt{2\pi} \quad \text{noting that } L > 0\end{aligned}$$

(b) **Method 1:**

For $t > 0$

$$\begin{aligned}0 &< \frac{1}{t+1} < 1 \\0 &< \frac{t^2}{t+1} < t^2 \\0 &< \frac{t^2 - 1 + 1}{t+1} < t^2 \\0 &< t - 1 + \frac{1}{t+1} < t^2 \\0 &< \int_0^x \left(t - 1 + \frac{1}{t+1} \right) dt < \int_0^x t^2 dt \\0 &< \ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}\end{aligned}$$

Method 2:

Let $f(x) = \ln(1+x) - x + \frac{x^2}{2}$ for $x > 0$

$$\begin{aligned}f'(x) &= \frac{1}{x+1} - 1 + x \\&= \frac{x^2}{x+1} \\&> 0 \quad \text{for } x > 0\end{aligned}$$

Since $f(0) = 0$ and $f(x)$ is increasing for $x > 0$ then $f(x) > 0$ in that domain. Hence

$$\ln(1+x) - x + \frac{x^2}{2} > 0$$

Let $g(x) = \ln(1+x) - x + \frac{x^2}{2} - \frac{x^3}{3}$ for $x > 0$

$$\begin{aligned}g'(x) &= \frac{x^2}{x+1} - x^2 \\&= \frac{x^2(1 - (x+1))}{x+1} \\&= -\frac{x^3}{x+1} \\&< 0 \quad \text{for } x > 0\end{aligned}$$

Since $g(0) = 0$ and $g(x)$ is decreasing for $x > 0$ then $g(x) < 0$ in that domain. Hence

$$\ln(1+x) - x + \frac{x^2}{2} - \frac{x^3}{3} < 0$$

Putting these inequalities together gives

$$0 < \ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}$$

- (c) (i) Let $F(x)$ and $G(z)$ be the cumulative distribution functions of random variables X and Z respectively

$$\begin{aligned}
 G(z) &= P(Z \leq z) \\
 &= P\left(\frac{X - n}{\sqrt{n}} \leq z\right) \\
 &= P(X \leq n + z\sqrt{n}) \\
 &= F(n + z\sqrt{n})
 \end{aligned}$$

Differentiate both sides with respect to z to get the probability density functions

$$g(z) = \sqrt{n} f(n + z\sqrt{n})$$

- (ii) Substituting part (i) into the probability density function for X

$$\begin{aligned}
 g(z) &= \sqrt{n} \times \frac{(n + z\sqrt{n})^n}{n!} e^{-(n+z\sqrt{n})} \\
 &= \frac{n^{n+\frac{1}{2}}}{e^n n!} \left(1 + \frac{z}{\sqrt{n}}\right)^n e^{-z\sqrt{n}}
 \end{aligned}$$

From part (b), let $x = \frac{z}{\sqrt{n}}$ where $z > 0$

$$\begin{aligned}
 0 &< \ln\left(1 + \frac{z}{\sqrt{n}}\right) - \frac{z}{\sqrt{n}} + \frac{z^2}{2n} < \frac{z^3}{3n\sqrt{n}} \\
 -\frac{z^2}{2} &< n \ln\left(1 + \frac{z}{\sqrt{n}}\right) - z\sqrt{n} < \frac{z^3}{3\sqrt{n}} - \frac{z^2}{2} \\
 e^{-\frac{z^2}{2}} &< \left(1 + \frac{z}{\sqrt{n}}\right)^n e^{-z\sqrt{n}} < e^{-\frac{z^2}{2} + \frac{z^3}{3\sqrt{n}}} \\
 \frac{n^{n+\frac{1}{2}}}{e^n n!} e^{-\frac{z^2}{2}} &< g(z) < \frac{n^{n+\frac{1}{2}}}{e^n n!} e^{-\frac{z^2}{2} + \frac{z^3}{3\sqrt{n}}}
 \end{aligned}$$

As $n \rightarrow \infty$ then $\frac{z^3}{3\sqrt{n}} \rightarrow 0$ and $\frac{n^{n+\frac{1}{2}}}{e^n n!} \rightarrow \frac{1}{\sqrt{2\pi}}$ from part (a)(v). Hence

$$\lim_{n \rightarrow \infty} g(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

This means that as the sample size n gets large, the distribution of Z approaches a standard normal distribution (in the positive domain).

Question 16

- (a) (i) From the horizontal displacement equation

$$\begin{aligned}x &= \frac{u^2}{g} \cos \theta \left(1 - e^{-\frac{gt}{u}}\right) \\x \sec \theta &= \frac{u^2}{g} \left(1 - e^{-\frac{gt}{u}}\right) \\e^{-\frac{gt}{u}} &= 1 - \frac{gx \sec \theta}{u^2} \\t &= -\frac{u}{g} \ln \left(1 - \frac{gx \sec \theta}{u^2}\right)\end{aligned}$$

Substitute into the vertical displacement equation

$$\begin{aligned}y &= \frac{u^2}{g} (1 + \sin \theta) \left(1 - e^{-\frac{gt}{u}}\right) - ut \\y &= x \sec \theta (1 + \sin \theta) + \frac{u^2}{g} \ln \left(1 - \frac{gx \sec \theta}{u^2}\right)\end{aligned}$$

At the range when $x = R$ then $y = 0$ hence

$$\begin{aligned}R \sec \theta (1 + \sin \theta) + \frac{u^2}{g} \ln \left(1 - \frac{gR \sec \theta}{u^2}\right) &= 0 \\ \ln \left(1 - \frac{gR \sec \theta}{u^2}\right) &= -\frac{gR}{u^2} (\sec \theta + \tan \theta)\end{aligned}$$

Let $f(\theta) = \frac{gR}{u^2} (\sec \theta + \tan \theta)$ which means that

$$\begin{aligned}\frac{gR \sec \theta}{u^2} &= 1 - e^{-f(\theta)} \\ R &= \frac{u^2}{g} \cos \theta (1 - e^{-f(\theta)})\end{aligned}$$

- (ii) Using the expression for the range in part (i)

$$\begin{aligned}R(\theta) &= \frac{u^2}{g} \cos \theta (1 - e^{-f(\theta)}) \\ R'(\theta) &= -\frac{u^2}{g} \sin \theta (1 - e^{-f(\theta)}) + \frac{u^2}{g} \cos \theta f'(\theta) e^{-f(\theta)}\end{aligned}$$

To remove the exponential terms, notice that by rearranging the equation of $R(\theta)$

$$\frac{u^2}{g} \cos \theta e^{-f(\theta)} = \frac{u^2}{g} \cos \theta - R(\theta)$$

Hence

$$R'(\theta) = -R(\theta) \tan \theta + f'(\theta) \left(\frac{u^2}{g} \cos \theta - R(\theta) \right)$$

Looking into $f'(\theta)$

$$\begin{aligned} f(\theta) &= \frac{gR(\theta)}{u^2} (\sec \theta + \tan \theta) \\ f'(\theta) &= \frac{g}{u^2} (R'(\theta)(\sec \theta + \tan \theta) + R(\theta)(\sec \theta \tan \theta + \sec^2 \theta)) \\ &= \frac{g}{u^2} (\sec \theta + \tan \theta)(R'(\theta) + R(\theta) \sec \theta) \end{aligned}$$

Substitute this into the derivative equation

$$R'(\theta) = -R(\theta) \tan \theta + \frac{g}{u^2} (\sec \theta + \tan \theta)(R'(\theta) + R(\theta) \sec \theta) \left(\frac{u^2}{g} \cos \theta - R(\theta) \right)$$

This simplifies to

$$R'(\theta) \left(\frac{gR(\theta)}{u^2} (\sec \theta + \tan \theta) - \sin \theta \right) = R(\theta) \sec \theta - \frac{g \sec \theta}{u^2} (\sec \theta + \tan \theta) [R(\theta)]^2$$

Hence

$$R'(\theta) = -\frac{R(\theta) \sec \theta \left(1 - \frac{gR(\theta)}{u^2} (\sec \theta + \tan \theta) \right)}{\frac{gR(\theta)}{u^2} (\sec \theta + \tan \theta) - \sin \theta}$$

A necessary condition for the maximum range is $R'(\theta) = 0$ so either

$$R(\theta) = 0 \quad \text{or} \quad R(\theta) = \frac{u^2}{g(\sec \theta + \tan \theta)}$$

Take the positive solution and substitute it back into the original equation

$$\begin{aligned} R(\theta) &= \frac{u^2}{g} \cos \theta (1 - e^{-f(\theta)}) \\ \frac{1}{\sec \theta + \tan \theta} &= \cos \theta (1 - e^{-f(\theta)}) \\ 1 - e^{-f(\theta)} &= \frac{\sec \theta}{\sec \theta + \tan \theta} \\ \frac{\sec \theta + \tan \theta}{\tan \theta} &= e^{f(\theta)} \\ \sin \theta &= \frac{1}{e^{f(\theta)} - 1} \end{aligned}$$

But when $R(\theta) = \frac{u^2}{g(\sec \theta + \tan \theta)}$ then $f(\theta) = 1$, hence $\sin \theta = \frac{1}{e - 1}$.

(iii) Using the expression for the first derivative in terms of $f(\theta)$

$$R'(\theta) = \frac{R(\theta) \sec \theta (1 - f(\theta))}{f(\theta) - \sin \theta}$$

$$R''(\theta) = \frac{g(\theta) - h(\theta)}{(f(\theta) - \sin \theta)^2}$$

Where

$$g(\theta) = \left\{ R'(\theta) [\sec \theta (1 - f(\theta))] + R(\theta) \frac{d}{d\theta} [\sec \theta (1 - f(\theta))] \right\} (f(\theta) - \sin \theta)$$

$$h(\theta) = R(\theta) \sec \theta (1 - f(\theta)) [f'(\theta) - \cos \theta]$$

When $\sin \theta = \frac{1}{e-1}$ then $R(\theta) = \frac{u^2}{g(\sec \theta + \tan \theta)}$, $R'(\theta) = 0$ and $f(\theta) = 1$ as shown in the previous part, so these simplify to

$$\begin{aligned} g(\theta) &= R(\theta)(1 - \sin \theta) \frac{d}{d\theta} [\sec \theta (1 - f(\theta))] \Big|_{f(\theta)=1} \\ &= R(\theta)(1 - \sin \theta) [\sec \theta \tan \theta (1 - f(\theta)) - f'(\theta) \sec \theta] \Big|_{f(\theta)=1} \\ &= -R(\theta)(1 - \sin \theta) f'(\theta) \sec \theta \\ h(\theta) &= 0 \end{aligned}$$

Hence at the stationary point

$$R''(\theta) = -\frac{R(\theta) f'(\theta) \sec \theta}{1 - \sin \theta}$$

At the stationary point $\sin \theta < 1$, $\sec \theta > 0$ and $R(\theta) > 0$. Also

$$\begin{aligned} f'(\theta) &= \frac{g}{u^2} (\sec \theta + \tan \theta) (R'(\theta) + R(\theta) \sec \theta) \\ &= \frac{gR(\theta)}{u^2} \sec \theta (\sec \theta + \tan \theta) \quad \text{as } R'(\theta) = 0 \\ &> 0 \end{aligned}$$

This means that $R''(\theta) < 0$ at the stationary point. Therefore, the range is maximised at $\theta = \sin^{-1} \left(\frac{1}{e-1} \right)$.

- (b) (i) Suppose that $\sqrt[3]{2}$ is rational, that is $\sqrt[3]{2} = \frac{p}{q}$ for some integers p and q , where p and q have no common factors and $q \neq 0$. Since $\sqrt[3]{4} = \frac{p^2}{q^2}$ is also a ratio of two integers with no common factors, then $\sqrt[3]{4}$ must also be rational. Hence, if $\sqrt[3]{2}$ is rational then $\sqrt[3]{4}$ is rational.

For the converse of the above, suppose that $\sqrt[3]{4}$ is rational, that is $\sqrt[3]{4} = \frac{a}{b}$ for some integers a and b , where a and b have no common factors and $b \neq 0$. Squaring both sides gives

$$2\sqrt[3]{2} = \frac{a^2}{b^2}$$

$$\sqrt[3]{2} = \frac{a^2}{2b^2} \quad \text{which is rational}$$

Hence, if $\sqrt[3]{4}$ is rational then $\sqrt[3]{2}$ is rational.

$\therefore \sqrt[3]{2}$ is rational if and only if $\sqrt[3]{4}$ is rational.

- (ii) If m is an integer where $m \in S_2$ then

$$m\sqrt[3]{2} = M \quad \text{where } M \text{ is an integer}$$

$$m^2\sqrt[3]{2} = Mm$$

Since Mm is an integer then $m^2 \in S_2$.

Squaring both sides of the first expression gives $m^2\sqrt[3]{4} = M^2$. Since M^2 is an integer then $m^2 \in S_4$.

Since $m^2 \in S_2$ and $m^2 \in S_4$ then $m^2 \in S$, so it follows that there is at least one positive integer in S .

- (iii) If k is an integer where $k \in S$, then

$$k\sqrt[3]{2} = P \quad \text{and} \quad k\sqrt[3]{4} = Q$$

where P and Q are integers.

First note that $[\sqrt[3]{4} - 1]k = Q - k$ which is an integer

Also $[\sqrt[3]{4} - 1]k \times \sqrt[3]{2} = 2k - P$ which is an integer so $[\sqrt[3]{4} - 1]k \in S_2$

Similarly $[\sqrt[3]{4} - 1]k \times \sqrt[3]{4} = 2P - Q$ which is an integer so $[\sqrt[3]{4} - 1]k \in S_4$

Hence, if $k \in S$, then $[\sqrt[3]{4} - 1]k \in S$.

- (iv) Since there can only be positive integers in S then there must exist a particular $k \in S$, which is the smallest positive integer in the set S . From the previous part $[\sqrt[3]{4} - 1]k \in S$ as well. However since

$$\begin{aligned} 1 &< 4 < 8 \\ \sqrt[3]{1} - 1 &< \sqrt[3]{4} - 1 < \sqrt[3]{8} - 1 \\ 0 &< [\sqrt[3]{4} - 1]k < k \end{aligned}$$

This implies that the set S contains a smaller integer than the minimum k . This is a contradiction.

Hence, it must be the case that the original statement, that $\sqrt[3]{2}$ is rational if and only if $\sqrt[3]{4}$ is rational, is not true.

Therefore, both $\sqrt[3]{2}$ and $\sqrt[3]{4}$ must be irrational.

Alternate Method: Using the same inequality it can be seen that

$$\begin{aligned} [\sqrt[3]{4} - 1]k &< k \\ [\sqrt[3]{4} - 1]^2 k &< [\sqrt[3]{4} - 1]k \\ [\sqrt[3]{4} - 1]^3 k &< [\sqrt[3]{4} - 1]^2 k \\ &\text{and so on} \dots \end{aligned}$$

This leads to an infinite sequence of strictly decreasing positive integers in S given by $\{k, [\sqrt[3]{4} - 1]k, [\sqrt[3]{4} - 1]^2 k, [\sqrt[3]{4} - 1]^3 k, \dots\}$.

However, there exists only finitely many positive integers below any given positive integer. By contradiction, it must be the case that the original statement, that $\sqrt[3]{2}$ is rational if and only if $\sqrt[3]{4}$ is rational, is not true.

Therefore, both $\sqrt[3]{2}$ and $\sqrt[3]{4}$ must be irrational.